

SKILLWEEK7

TASK1: Parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs

```
hrushika_habeeb@Hrushika-Habeeb: ~/
+ -
hrushika_habeeb@Hrushika-Habeeb:~$ mkdir Skill7
hrushika_habeeb@Hrushika-Habeeb:~$ mkdir Skill7
hrushika_habeeb@Hrushika-Habeeb:~$ cd Skill7
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ pwd
/home/hrushika_habeeb/Skill7
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ >
-bash: syntax error near unexpected token `newline'
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ echo "Hello Ubuntu" > input.txt
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ cat input.txt
Hello Ubuntu
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ cat
h
h

exit
exit
>
>
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ cat < input.txt
Hello Ubuntu
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ cat < missing.txt
-bash: missing.txt: No such file or directory
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ ls
input.txt
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ echo "This is another test"
> test.txt
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ cat < test.txt
This is another test
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ |
```

TASK2: Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection

```
hrushika_habeeb@Hrushika-Habeeb: ~/ X + -
-bash: syntax error near unexpected token `newline'
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ echo "Hello from Ubuntu" > output.txt
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ cat output.txt
Hello from Ubuntu
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ echo "Hello from Ubuntu" > output.txt
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ ls -l output.txt
-rw-r--r-- 1 hrushika_habeeb hrushika_habeeb 18 Sep 18 16:35 output.txt
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ cat output.txt
Hello from Ubuntu
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ echo "First line" > output.txt
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ echo "Second line" > output.txt
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ cat output.txt
Second line
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ echo "Test" > permission.txt
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ chmod 444 permission.txt
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ ls -l permission.txt
-r--r--r-- 1 hrushika_habeeb hrushika_habeeb 5 Sep 18 16:36 permission.txt
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ echo "New data" > permission.txt
-bash: permission.txt: Permission denied
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ chmod 644 permission.txt
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ echo "New data" > permission.txt
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ cat permission.txt
New data
hrushika_habeeb@Hrushika-Habeeb:~/Skill7$ |
```


TASK3: Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios

```
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ echo "Line 1" > append.txt
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ cat append.txt
Line 1
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ echo "Line 2" >> append.txt
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ cat append.txt
Line 1
Line 2
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ echo "Line 3" >> append.txt
echo "Line 4" >> append.txt
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ cat append.txt
Line 1
Line 2
Line 3
Line 4
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ echo "New file" > example.txt
t
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ echo "Added line" >> example
.txt
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ cat example.txt
New file
Added line
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ echo "Replaced" > example.tx
t
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ cat example.txt
Replaced
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ |
```

TASK4: To Parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases

```
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ ls nonexistent.txt
ls: cannot access 'nonexistent.txt': No such file or directory
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ ls nonexistent.txt 2> error.txt
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ cat error.txt
ls: cannot access 'nonexistent.txt': No such file or directory
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ ls Skill7 2> error.txt
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ cat error.txt
ls: cannot access 'Skill7': No such file or directory
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ ls existing.txt > output.txt
ls: cannot access 'existing.txt': No such file or directory
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ ls missing.txt 2> error.txt
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ command > output.txt 2>&1
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ ls existing.txt missing.txt > result.txt 2>&1
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ cat result.txt
ls: cannot access 'existing.txt': No such file or directory
ls: cannot access 'missing.txt': No such file or directory
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ ls existing.txt missing.txt &> result.txt
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ echo "Hello" > output.txt
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ echo "Hello" > output.txt
echo "This appears on screen"
This appears on screen
Hrushika_Habeeb@Hrushika-Habeeb:~/Skill7$ |
```

